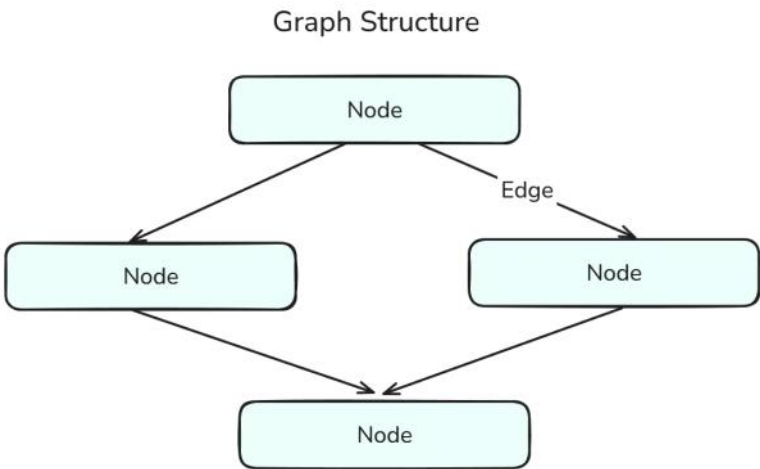
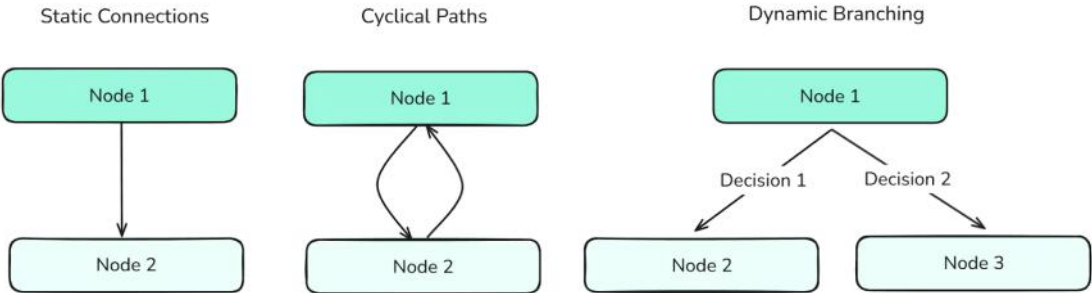
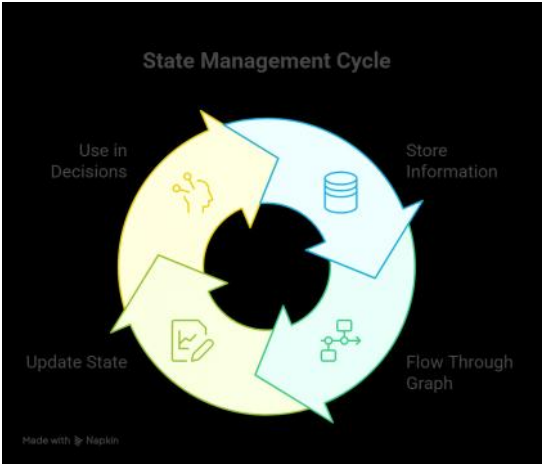
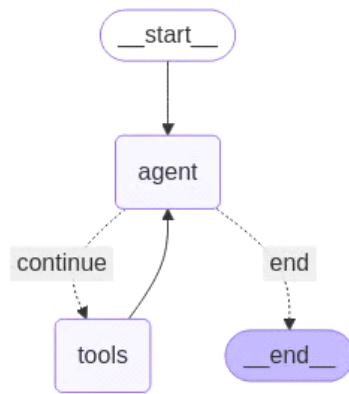






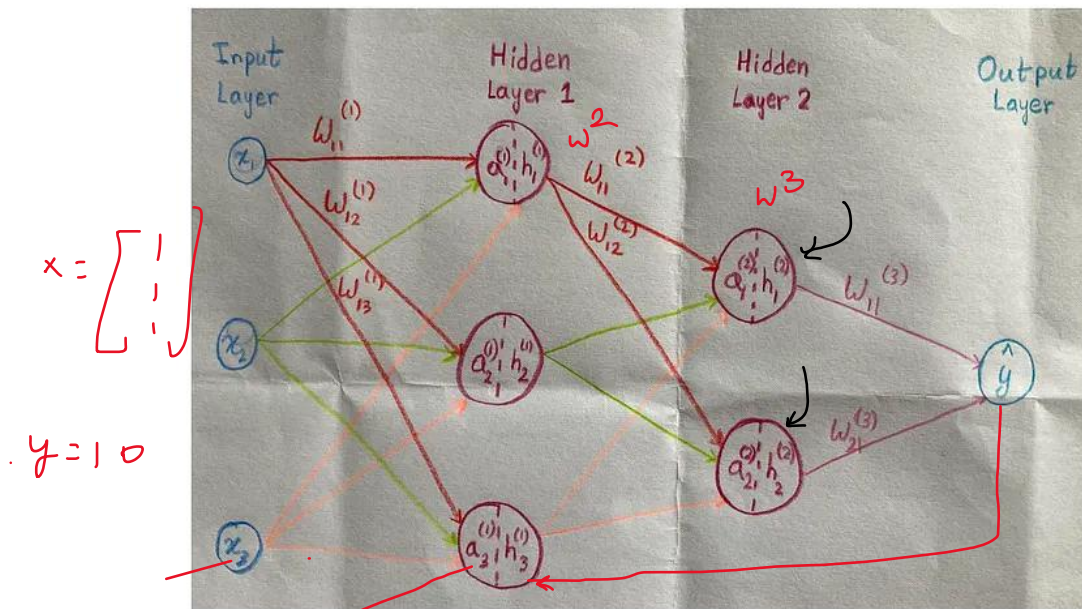
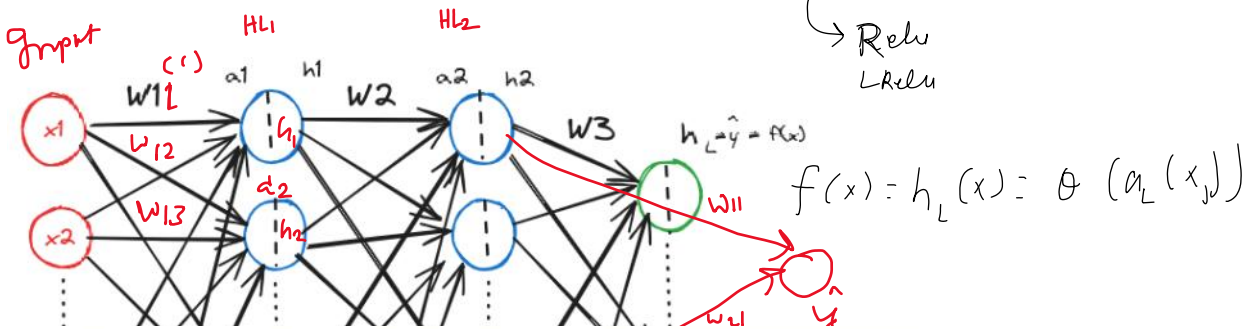
$$L = \frac{1}{2} (y - \hat{y})^2$$

$\frac{\partial L}{\partial w}$

$$a_i(x) = b_i + w_i h_{i-1}(x)$$

$$h_i(x) = g(a_i(x))$$

ReLU
LReLU



test image

$$W^{(1)} = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

$$W^{(2)} = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix}$$

$$W^{(3)} = [2, 5]$$

Figure 2: Feedforward Neural Network

This neural network does not contain any biases and the weights of the network are given as follows:

$$W^{(1)} = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \quad W^{(2)} = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} \quad W^{(3)} = [2 \quad 5]$$

The input to the network is

[1]

The input to the network is

$$X = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

The target value of y is $y = 10$

Notation: The superscript (i) represents layer i and subscript (i, j) represents input node i to output node j .

Step 1:

$$a^{(1)} = w^{(1)} \times X = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$h^{(1)} = \sigma(a^{(1)}) = \begin{bmatrix} \frac{1}{1+e^{-4}} \\ \frac{1}{1+e^{-5}} \\ \frac{1}{1+e^{-6}} \end{bmatrix} = \begin{bmatrix} 0.982 \\ 0.993 \\ 0.997 \end{bmatrix}$$

$$a^{(2)} = w^{(2)} \times h^{(1)} = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0.982 \\ 0.993 \\ 0.997 \end{bmatrix} = \begin{bmatrix} 3.969 \\ 4.936 \end{bmatrix}$$

$$h^{(2)} = \sigma(a^{(2)}) = \begin{bmatrix} \frac{1}{1+e^{-3.969}} \\ \frac{1}{1+e^{-4.936}} \end{bmatrix} = \begin{bmatrix} 0.981 \\ 0.992 \end{bmatrix}$$

$$y = w^{(3)} h^{(2)} = [2 \ 5] \begin{bmatrix} 0.981 \\ 0.992 \end{bmatrix} = 6.927$$

$$L = (p_y - t_y)^2 = (6.927 - 10)^2 = 9.443$$

$$\frac{dL}{d\hat{y}} = \frac{d}{d\hat{y}} (\hat{y} - y)^2$$

$$\frac{2(\hat{y} - y)}{2(6.927 - 10)} = \boxed{-6.15}$$

$$\nabla w^{(3)} = \begin{bmatrix} \frac{\partial L}{\partial w_{11}^{(3)}} & \frac{\partial L}{\partial w_{21}^{(3)}} \end{bmatrix}^T$$

$$\frac{\delta L}{\delta w_{11}^{(2)}} = \frac{\delta L}{\delta y} \times \frac{\delta y}{\delta w_{11}^{(2)}}$$

$$\frac{\delta L}{\delta y} \times \frac{\delta}{\delta w_{11}^{(2)}} (w_{11}^{(2)} h_1^{(2)} + b_{21}^{(2)} h_2^{(2)})$$

$$\frac{\delta L}{\delta y} \times h_1^{(2)}$$

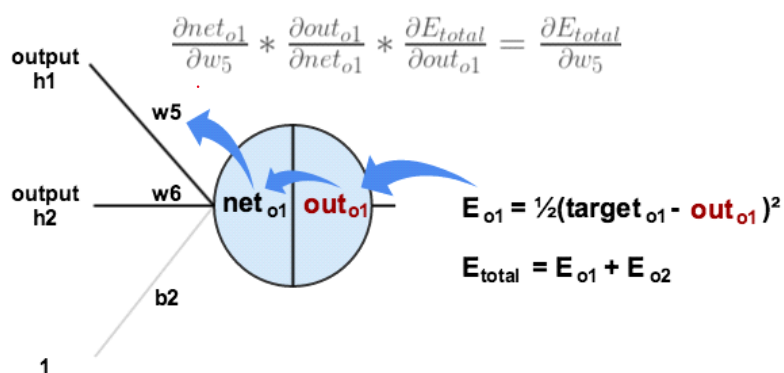
$$-6.156 \times 0.991 = -6.039$$

SL.

$$\frac{\delta L}{\delta w_{21}^{(2)}} = \frac{\delta L}{\delta y} \times h_2^{(2)}$$

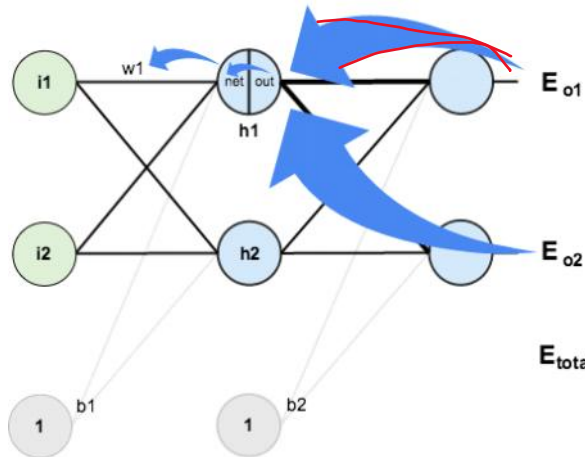
$$= -6.106$$

$$\nabla W^{(2)} = \begin{bmatrix} \frac{\delta L}{\delta w_{11}^{(2)}} & \frac{\delta L}{\delta w_{21}^{(2)}} & \frac{\delta L}{\delta w_{31}^{(2)}} & \frac{\delta L}{\delta w_{41}^{(2)}} & \frac{\delta L}{\delta w_{51}^{(2)}} & \frac{\delta L}{\delta w_{61}^{(2)}} \end{bmatrix}^T$$

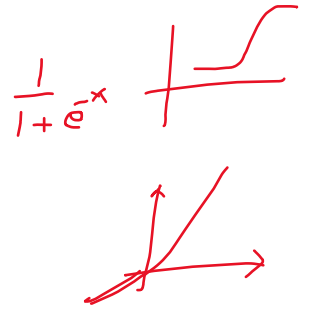


$$\frac{\partial E_{total}}{\partial w_1} = \frac{\partial E_{total}}{\partial out_{h1}} * \frac{\partial out_{h1}}{\partial nct_{h1}} * \frac{\partial nct_{h1}}{\partial w_1}$$

$$\frac{\partial E_{total}}{\partial out_{h1}} = \frac{\partial E_{o1}}{\partial out_{h1}} + \frac{\partial E_{o2}}{\partial out_{h1}}$$



f 50% 50% 50%



Signal

